Evaluating Treatment Prioritization Rules via Rank-Weighted Average Treatment Effects

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Background Question

Given a score that **ranks people** for treatment, does it put those who **benefit most** at the top?

Classification

- Score ranks people by *risk*
- Evaluate the ranking with **AUROC**
- No threshold needed

Treatment prioritization

- Score ranks people by *expected benefit*
- Evaluate the ranking with **RATE**
- No threshold needed

RATE: **R**ank-weighted **A**verage **T**reatment **E**ffect

Why Is This Needed?

Two common ways to build a prioritization rule

- **CATE-based**: estimate each person's treatment effect and treat the largest first.
  Principled, but needs a lot of trial data.
- **Risk-based**: treat the people at highest baseline risk first.
  A heuristic, but risk models are easy to build and widely used.

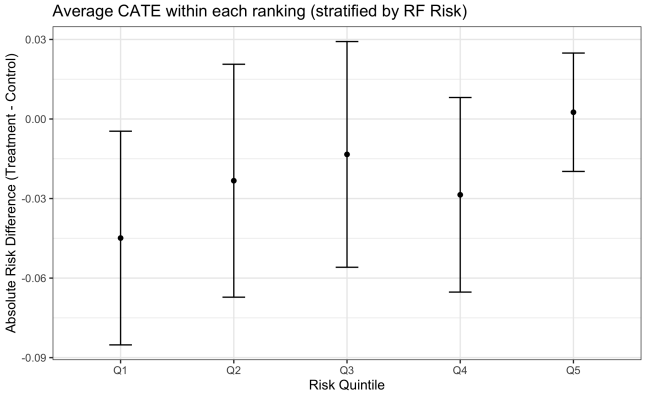
Why not just evaluate a treatment policy?

- In practice the cut-off (who actually gets treated) depends on cost, capacity, side effects, preferences.
- It is often **unknown** at the time we evaluate the rule.

Goal: a single metric that works for any rule, needs no threshold, and comes with valid confidence intervals.

Motivating Example: Aspirin for Acute Stroke

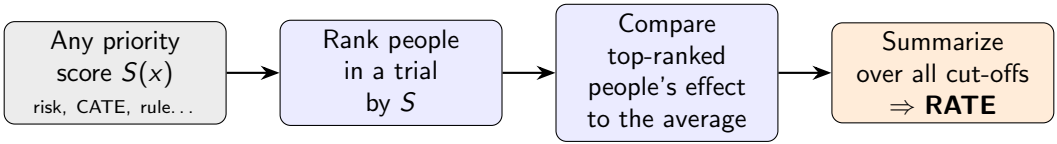
- Aspirin is recommended for almost all stroke patients, but it also raises bleeding risk.
- **Are there patients who benefit less?**
- International Stroke Trial: ∼19,000 patients.
- Ranking by risk shows a trend, but no subgroup is significant.



Treatment effect by risk quintile (lower = more benefit).

Can we test the *whole ranking at once*, without arbitrary subgroups?

Big Picture of the Main Idea



- If S ranks well, top-ranked people should benefit **more than average**. RATE measures exactly this, regardless of how S was built.
- RATE $= 0$: the ranking is no better than random.

Knowledge Background

Three ingredients: potential outcomes define what we want, CATE is the target, and AUROC motivates the approach.

- **Potential outcomes**: each person has $Y(1)$ (treated) and $Y(0)$ (untreated); we only observe one.
- **CATE**: average treatment effect for people with covariates x

$$\tau(x) = \mathbb{E}\big[Y(1) - Y(0) \mid X = x\big]$$

- **AUROC**: the analogy for threshold-free evaluation of a ranking.

Scores for Individual Effects: IPW and AIPW

We want $Y_i(1) - Y_i(0)$, but only one of the two is observed.
Idea: build a score Γ_i that equals the treatment effect *on average*, $\mathbb{E}[\Gamma_i \mid X_i] = \tau(X_i)$.

IPW score (simple fix, π = probability of treatment):

$$\Gamma_i^{\text{IPW}} = \frac{\mathbf{1}(W_i = 1) \cdot Y_i(1)}{\pi} - \frac{\mathbf{1}(W_i = 0) \cdot Y_i(0)}{1 - \pi}$$

Unbiased, but noisy: it uses the raw outcomes.

AIPW score (better):

$$\hat{\Gamma}_i = \underbrace{\hat{m}(X_i, 1) - \hat{m}(X_i, 0)}_{\text{outcome-model prediction}} + \underbrace{\frac{W_i - \hat{e}(X_i)}{\hat{e}(X_i)\,(1 - \hat{e}(X_i))}\big(Y_i - \hat{m}(X_i, W_i)\big)}_{\text{IPW correction on residuals}}$$

Doubly robust: valid if either the outcome model $\hat{m}$ or the propensity model $\hat{e}$ is correct.
Lower variance: the outcome model absorbs baseline variation, so what is left is a much smaller residual.

Setup

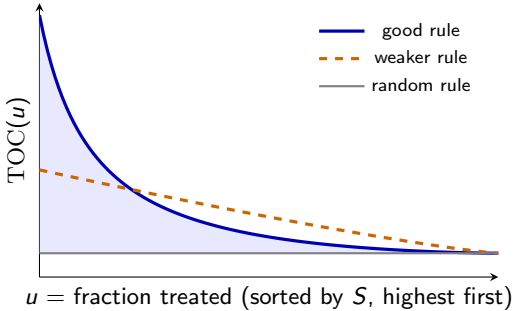
- A **prioritization rule** is a score $S(x)$: higher score $\Rightarrow$ treat first.
- S is **fixed** and built elsewhere. We only *evaluate* it on a separate test set (ideally a trial).
- Each person's **priority quantile** $F_S(S(X_i)) \in (0, 1)$:
  ≈ 1 means ranked at the very top.

We now build the TOC curve on top of this.

Step 1: The TOC Curve

$$\mathrm{TOC}(u) = \underbrace{\mathbb{E}\big[Y_i(1) - Y_i(0) \mid F_S(S(X_i)) \geq 1 - u\big]}_{\text{effect among the top } u \text{ fraction by } S} - \underbrace{\mathbb{E}\big[Y_i(1) - Y_i(0)\big]}_{\text{overall effect}}$$

- "Targeting Operator Characteristic", an analogue of the ROC curve.
- Random ranking: $\mathrm{TOC} = 0$ everywhere.
- Good ranking: high at small u, falls to 0 at $u = 1$.



Step 2: RATE = Weighted Area Under the TOC

$$\text{RATE} = \int_0^1 \alpha(u)\,\text{TOC}(u)\,du$$

Different weights $\alpha(u)$ give a whole family. Two main members:

Metric	Weight	Reads as
AUTOC	$\alpha(u) = 1$	plain area under the TOC curve
Qini coefficient	$\alpha(u) = u$	area under the cumulative-gain curve

- **AUTOC**: proposed by Zhao et al. (2013) in the statistics literature.
- **Qini coefficient**: proposed by Radcliffe (2007) in the marketing literature.
- This paper shows they are **special cases of one framework**, which gives unified inference for the first time.

Another View: RATE Is a Weighted Treatment Effect

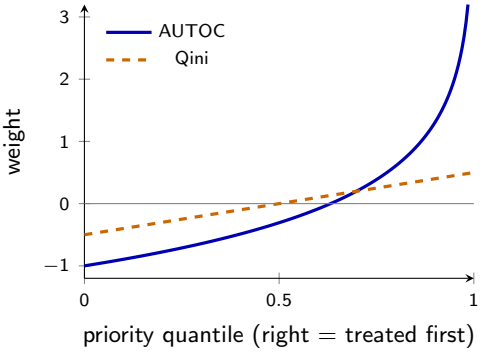
Any RATE can be rewritten as

$$\text{RATE} = \mathbb{E}\big[\, w(\text{rank}_i) \cdot (Y_i(1) - Y_i(0)) \,\big]$$

with weights that depend only on the rank.

- **AUTOC**: very large weight on the top few people.
- **Qini**: weights spread evenly across the ranking.
- Weights average to zero $\Rightarrow$ RATE $= 0$ when the ranking is uninformative, so "RATE $= 0$" is a natural null hypothesis.

Why it matters: once we replace $Y_i(1) - Y_i(0)$ with $\hat{\Gamma}_i$, RATE is just a weighted sum of numbers we can actually compute. That is what makes the CLT possible.



Step 3: Estimating It from Data

Problem: individual effects $Y_i(1) - Y_i(0)$ are never observed.
Fix: replace them with the AIPW scores $\hat{\Gamma}_i$ from before, which are correct *on average*.

Estimate

1. Sort the test set by S; $i(j)$ is the person ranked j-th.
2. Mean of $\hat{\Gamma}$ in the top u, minus the overall mean:

$$\widehat{\mathrm{TOC}}(u) = \frac{1}{\lceil un \rceil} \sum_{j=1}^{\lceil un \rceil} \hat{\Gamma}_{i(j)} - \frac{1}{n} \sum_{i=1}^{n} \hat{\Gamma}_i$$

3. Average over cut-offs with the chosen weights:

$$\widehat{\mathrm{RATE}} = \frac{1}{n} \sum_{j=1}^{n} \alpha\Big(\frac{j}{n}\Big) \widehat{\mathrm{TOC}}\Big(\frac{j}{n}\Big)$$

Same recipe for RCTs, observational studies, and survival outcomes; only $\hat{\Gamma}_i$ changes.

Theoretical Results: Valid Confidence Intervals

Theorem 3 (central limit theorem, informally)

Under regularity conditions, $\widehat{\text{RATE}}$ is approximately normal around the true RATE.

Why is this hard? The estimator combines two structures at once:

- **L-statistic**: the weights come from *estimated* ranks (order statistics).
- **Doubly-robust estimator**: AIPW scores with estimated nuisance models.

Merging these two asymptotic theories is the technical core of the paper. It yields the first CLT for Qini and AUTOC, even in a simple RCT.

- **In practice**: estimate the standard error by recomputing RATE on random half-samples (**half-sample bootstrap**).

What RATE Can Do

1. **Is my rule better than random?**
   Test RATE $= 0$.

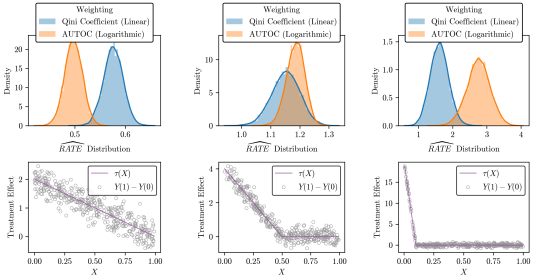
2. **Is rule A better than rule B?**
   Test the paired difference (e.g. risk score vs. CATE model).

3. **Is there any predictable treatment-effect heterogeneity?**
   Use a flexible CATE model as the rule and test RATE $= 0$.

Properties

- **Model-agnostic**: risk scores and CATE models are judged the same way.
- **Threshold-free**: summarizes all cut-offs at once.
- **General**: works for binary, continuous, and survival outcomes.

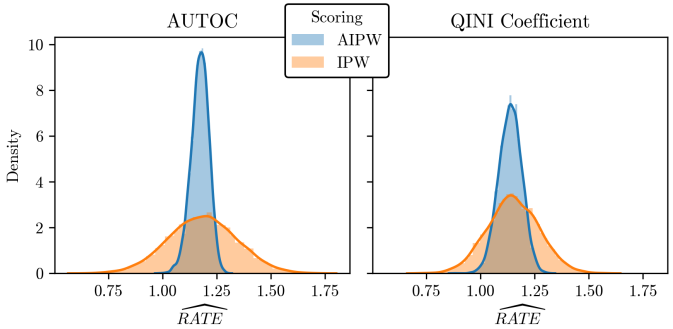
Which Weight? AUTOC vs. Qini



Simulation: share of people with a non-zero effect = 100% / 50% / 10% (left to right).

- Effect spread across everyone ⇒ **Qini** more powerful.
- Effect concentrated in a few ⇒ **AUTOC** more powerful.
- Why: AUTOC puts much larger weight on the top-ranked people, which is exactly where the signal is when only a few benefit.

Which Score? IPW vs. AIPW



- Both are centered at the truth.
- Adjusting for covariates (AIPW) gives **much tighter** estimates, even in an RCT.
- Why: AIPW uses residuals $Y_i - \hat{m}(X_i, W_i)$ instead of raw Y_i. The outcome model absorbs the baseline variation, leaving a much smaller noise term.

Study 1: Aspirin and Stroke (International Stroke Trial, IST)

Hypothesis: high-risk patients benefit *less* from aspirin. Rules trained on half of the trial, evaluated on the other half.

Prioritization rule	AUTOC (95% CI)	p
Random forest risk score	**0.014 (0.002, 0.027)**	**0.022**
Logistic regression risk score	0.011 (0.000, 0.230)	0.057
Causal forest (CATE)	0.006 (−0.012, 0.024)	0.49
Logistic T-learner (CATE)	0.005 (−0.013, 0.023)	0.59

- One test on the whole ranking finds an effect that quintile subgroups missed.
- Here the simple **risk score beats the CATE models**.
- Why: when benefit tracks baseline risk, a risk score already ranks well. Risk is a single outcome and is easier to learn, while a CATE is a difference of two noisy outcomes.

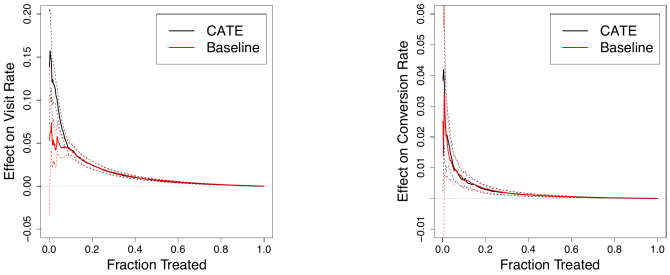
Study 2: Intensive Blood-Pressure Control

- Two trials: **SPRINT** (significant benefit) and **ACCORD-BP** (not significant). The literature suggested the difference is due to heterogeneous treatment effects (HTE).
- Rules trained on one trial, evaluated on the other; survival outcome.

	Evaluated on SPRINT		Evaluated on ACCORD-BP	
Prioritization rule	AUTOC (95% CI)	p	AUTOC (95% CI)	p
Causal survival forest (CATE)	2.37 (−3.52, 8.27)	0.43	−5.29 (−17.58, 7.00)	0.40
Cox PH S-learner (CATE)	3.57 (−1.92, 9.07)	0.20	8.90 (−2.62, 20.43)	0.13
Random survival forest risk	4.21 (−2.46, 10.88)	0.22	2.93 (−8.82, 14.67)	0.62
Framingham risk score	1.83 (−4.28, 7.94)	0.55	−7.01 (−18.45, 4.48)	0.25
ACC/AHA pooled cohort equations	−1.76 (−8.26, 4.73)	0.59	−1.43 (−13.22, 10.36)	0.81

- **No rule**, risk-based or CATE-based, ranks patients by benefit significantly better than random, on either trial.
- Neither trial is powered enough for a conclusive claim of heterogeneity (survival-time scale), so the HTE explanation is not supported.
- No evidence that risk calculators like Framingham pick out who benefits.

Study 3: Online Advertising (Criteo, 25M users)

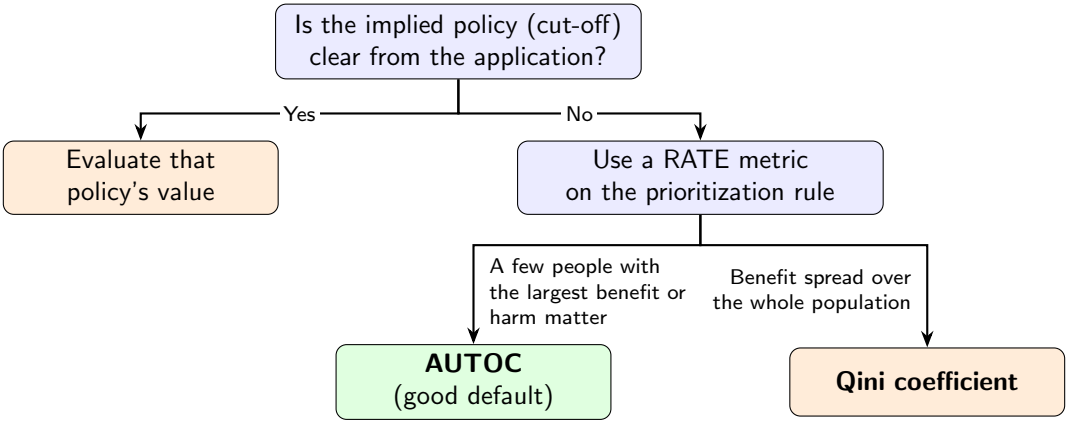


TOC curves: CATE model vs. baseline-risk rule.

Question: show an ad to users with the largest predicted uplift (CATE), or to users likely to visit anyway (baseline)? Data from randomized ad experiments.

- Both rules are better than random.
- CATE rule is better **only within the top ∼10%**.
- AUTOC detects it (+25%); **Qini does not**.
- The choice of weight changes the conclusion.

Authors' Recommendation



- If the cut-off is known, just evaluate that policy.
- If not, use RATE, with **AUTOC as the default**.

Discussion

Strengths

- Simple and intuitive.
- One framework that unifies existing metrics, with valid inference.
- Risk scores and CATE models are judged by the same standard.

Limitations

- Evaluates the *ranking* only, not the size of predicted effects or the best cut-off.
- Still needs trial (or unconfounded) data to evaluate.
- Sample splitting costs power.
- The direction (who goes to the top) and weight (AUTOC vs Qini) must be pre-specified.